1.10 Quadratics: The Discriminant

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Starter

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How many **distinct** real solutions do each of the following have?

$x^2 - 12x + 36 = 0$
$x^2 + x + 3 = 0$
$x^2 - 2x - 1 = 0$

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$$ax^2 + bx + c = 0$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note: Roots of a **function** f are the values of x such that $f(x) = 0$.
Similarly the roots of an **equation** are solutions to an equation in the form $f(x) = 0$

Looking at this formula, when in general do you think we have:

- No real roots?
- Equal roots?
- Two distinct roots?

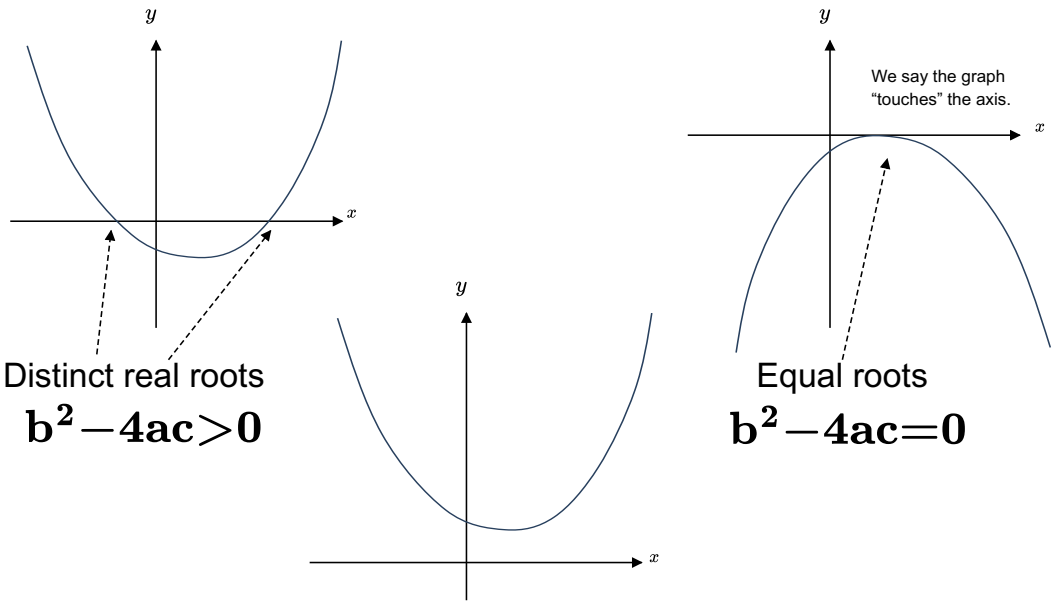


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Just for your interest…

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Why do we say "Equal Roots" not "One root"?

$$x^2 - 12x + 36 = 0$$

Using the quadratic formula gives us the same value in both $+$ and $-$ cases: $x = 6$.

You might wonder why we say "*it has one repeated root*" or "*it has equal roots*", i.e. indicating we have **2 roots** (but with the same value). Why not say it has 1 root?

It is due to the **Fundamental Theorem of Algebra**:

"Every polynomial of order n has exactly n roots."

A **polynomial** is an expression with non-negative integer powers of x, i.e. $a + bx + cx^2 + dx^3 + \ldots$ All linear, quadratic and cubic expressions are examples of polynomials.

The **order** of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

These roots might be repeated or might not be 'real' roots. $\sqrt{-1}$ is known as a **complex number**, which you will encounter if you do FM. But **it is still a value**!

The theorem means that a quadratic (order 2) will **always** have 2 roots. This is why you should say "no **real** roots" when $b^2 - 4ac < 0$ rather than "no roots", because there are still roots – it's just they're not 'real'! Similarly we must say "*equal roots*" because there are still 2 roots.

There are various other 'Fundamental Laws'. The '**FL of Arithmetic**' you encountered at KS3, which states that "*every positive integer* > 1 *can be written as a product of primes in one way only*". You will encounter the '**FL of Calculus**' in Chapter 13.

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by **Argand**. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is **provably** no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

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Quickfire Questions

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Equation	Discriminant	Number of Distinct Real Roots
$x^2 + 3x + 4 = 0$		
$x^2 - 4x + 1 = 0$		
$x^2 - 4x + 4 = 0$		
$2x^2 - 6x - 3 = 0$		
$x - 4 - 3x^2 = 0$		
$1 - x^2 = 0$		

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Problems involving the discriminant

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8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

a)

Tip: Always start by writing out a, b and c explicitly.

b)

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2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

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4 Given that the function $\mathrm{f}(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

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6 The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.
a Find the value of p.
b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

7 $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.
a Find the discriminant of $h(x)$ in terms of k.
b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k.

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Test Your Understanding

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$$x^2 + 5kx + (10k + 5) = 0$$

where k is a constant.
Given that this equation has equal roots, determine the value of k.

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

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Extension Questions:

1. [MAT 2009 1C] Given a real constant c, the equation
$$x^4 = (x-c)^2$$
Has four real solutions (including possible repeated roots) for:
A) $c \leq \frac{1}{4}$
B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$
C) $c \leq -\frac{1}{4}$
D) all values of c

2. [MAT 2006 1B] The equation
$\left(2 + x - x^2\right)^2 = 16$ has how many real root(s)?

3. [MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:
A) $P^3 > A$
B) $A^2 > 2P + 1$
C) $P^2 \geq 16A$
D) $PA > A + P$